AITS
ALL INDIA TEST SERIES
SELECTION HOGA YAHIN SE

Prayas JEE AIR 2025

Dropper JEE Main

AJPM6/05

Weekly Test 30 (AITS 05)

DURATION : 180 Minutes | **DATE : 12/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

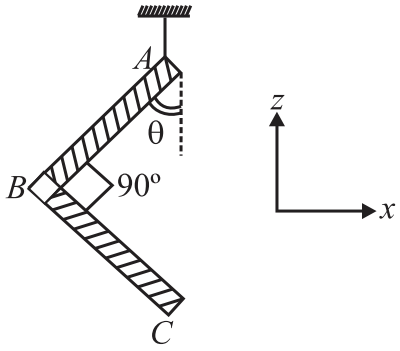
**** Use above values unless otherwise specified in a question.***

❑ ❑ ❑

SECTION-I (PHYSICS)

Single Correct Type Questions

1. An *L*-shaped object, made of two thin rods of same mass and uniform mass densities, is suspended with a string as shown in figure. If *BC* = 2*AB*, and the angle made by *AB* with downward vertical is θ, then:



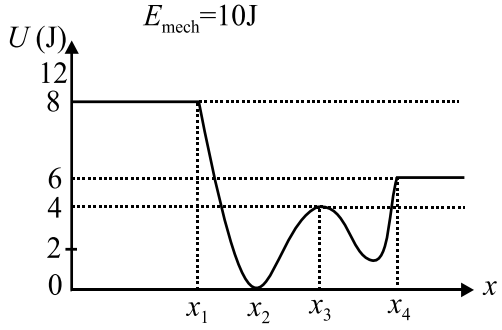
(1) $\tan\theta = \frac{1}{2\sqrt{3}}$

(2) $\tan\theta = \frac{1}{3}$

(3) $\tan\theta = \frac{2}{\sqrt{3}}$

(4) $\tan\theta = \frac{2}{3}$

2. Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion along *x*-axis, while only a conservative force $F(x)$ acts on it. Suppose that total mechanical energy E_{mech} = 10 J, the incorrect statement for this system is:



[where K.E. = kinetic energy]

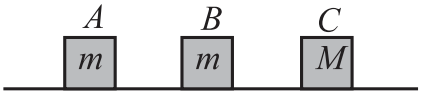
(1) at $x = x_3$, *K.E.* = 4 J

(2) at $x > x_4$, *K.E.* is constant throughout the region

(3) Among the shown positions, *K.E.* is smallest and the particle is moving at the slowest speed at $x < x_1$

(4) Among the shown positions, *K.E.* is greatest and the particle is moving at the fastest speed at $x = x_2$

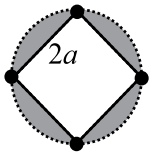
3. Three blocks *A*, *B* and *C* are lying on a smooth horizontal surface, as shown in the figure. *A* and *B* have equal masses *m*, while *C* has mass *M*. Block *A* is given an initial speed *v* towards *B* due to which it collides with *B* perfectly inelastically. The combined mass collides with *C*, also perfectly inelastically. If $\frac{3}{4}$th of the initial kinetic energy is lost in whole process. What is value of *M*/*m*?



(1) 5 (2) 2

(3) 4 (4) 3

4. Four identical particles of mass *M* are located at the corners of a square of side '2*a*'. What should be their speed if each of them revolves under the influence of other's gravitational field in a circular orbit circumscribing the square?



(1) $1.21\sqrt{\frac{GM}{a}}$

(2) $1.41\sqrt{\frac{GM}{a}}$

(3) $0.82\sqrt{\frac{GM}{a}}$

(4) $1.35\sqrt{\frac{GM}{a}}$

5. When M_1 gram of ice at –5° C (specific heat = 0.5 cal g^{-1} °C^{-1}) is added to M_2 gram of water at 50°C (specific heat = 1 cal g^{-1} °C^{-1}), finally no ice is left and the water is at 0°C. The value of latent heat of ice, in cal g^{-1} is:

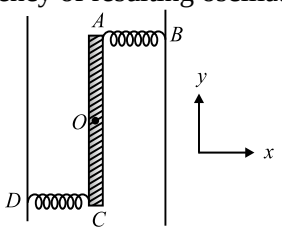
(1) $\frac{5M_1}{M_2} - 50$

(2) $\frac{50M_2}{M_1}$

(3) $\frac{50M_2}{M_1} - 2.5$

(4) $\frac{50M_2}{M_1} - 5$

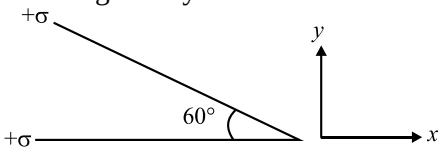
6. A spring of spring constant k is cut into two equal half AB & CD and are attached horizontally at the two ends of a uniform horizontal rod AC of length l and mass m as shown. The rod is pivoted at its centre 'O' and can rotate freely in horizontal plane. The other ends of the two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is:



(1) $\frac{1}{\pi}\sqrt{\frac{3k}{m}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{6k}{m}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$

7. Two infinite (non-conducting) planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 60°. The electric field in the region shown between them is given by:



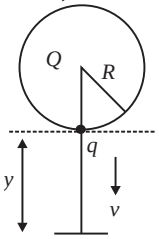
(1) $\frac{\sigma}{2\varepsilon_0}\left|\left(-\frac{\sqrt{3}}{2}\right)\hat{x}+\frac{\hat{y}}{2}\right|$

(2) $\frac{\sigma}{2\varepsilon_0}\left|\left(1+\sqrt{3}\right)\hat{y}-\frac{\hat{x}}{2}\right|$

(3) $\frac{\sigma}{2\varepsilon_0}\left|\left(1+\sqrt{3}\right)\hat{y}+\frac{\hat{x}}{2}\right|$

(4) $\frac{\sigma}{\varepsilon_0}\left|\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{\hat{x}}{2}\right|$

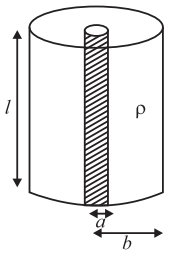
8. A solid sphere of radius R = 10 m carries a charge Q = 1 mC distributed uniformly over its volume. A very small point like piece of mass m = 1 gm gets detached from the bottom of the sphere and falls down vertically under gravity and mutual repulsion. This piece carries charge q = 1 μC. If it acquires a speed v when it has fallen through a vertical height y = 0.5 m (use $y \lll R$). Then (Assume the remaining portion to be spherical and take g = 10 m/s^2)



(1) v = 5 m/sec. (2) v = 10 m/sec.

(3) v = 20 m/sec. (4) v = 40 m/sec.

9. Model a torch battery of length l to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' (b = 3a) filled in between, with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R, the maximum Joule heating in R will take place for



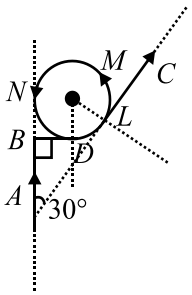
(1) $R=\frac{\rho}{\pi l}\ln(3)$

(2) $R=\frac{2\rho}{\pi l}\ln(2)$

(3) $R=\frac{\rho}{2\pi l}\ln(3)$

(4) $R=\frac{\rho}{2\pi l}\ln(2)$

10. A very long insulated metallic wire $ABDMNDLC$ is carrying current I, AB and LC parts are straight, long and at an angle of 30° as shown in figure. At D wire forms one circular turn $DMND$ of radius R and further an arc till L. AB and LC parts are tangential to circular turn at N and L. Magnetic field at the centre of circle is



(1) $\frac{\mu_0 I}{2R}$ (2) $\frac{\mu_0 I}{4\pi R}\left(\sqrt{2}+\frac{7\pi}{3}\right)$

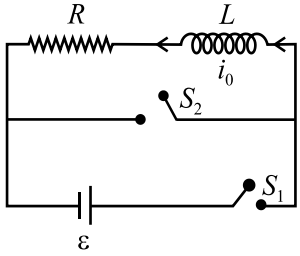
(3) $\frac{\mu_0 I}{2\pi R}(\pi+1)$ (4) $\frac{\mu_0 I}{2\pi R}\left(\pi+\frac{1}{\sqrt{2}}\right)$

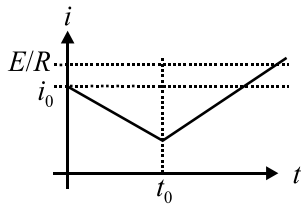
11. The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 75%. The percentage change in voltage sensitivity of the galvanometer will be:

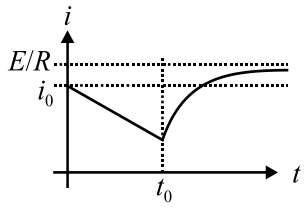
(1) 100% (2) 50%

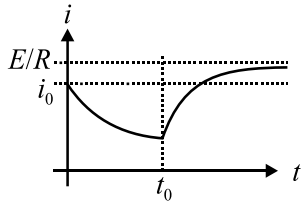
(3) 75% (4) 0%

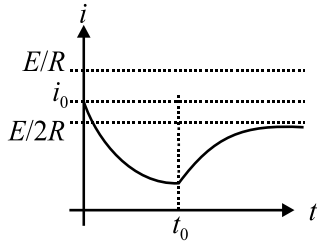
12. Initially, there is some current in the inductor i_0 in the circuit shown. The switch S_2 is closed at time $t = 0$ and the switch S_1 is kept open. At some later time (t_0) the switch S_2 is opened and S_1 is closed. The behaviour of the current in the inductor i as a function of time 't' is (Given $i_0 < E/R$)



(1)


(2)


(3)


(4)


13. A ball-1 of mass m is thrown vertically upward. Another ball-2 of mass $2m$ is thrown at angle θ with the vertical such that by the time ball-1 comes to the ground ball-2 is at its highest point. The ratio of the heights attained by the two balls respectively is $\frac{1}{x}$. The value of x is _____.

(1) 8 (2) 2
(3) 16 (4) 4

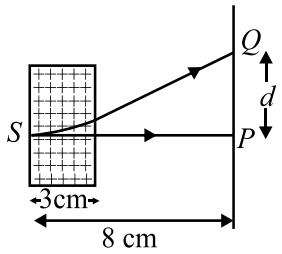
14. Electric field of a plane electromagnetic wave propagating through a non-magnetic medium is given by $E = 30\cos(2 \times 10^{10}\, t - 100x)$V/m. The dielectric constant of the medium is equal to: (Take $\mu_r = 1$)

(1) 2.25
(2) 1.15
(3) 1.5
(4) 9

15. An AC circuit has an inductor and a resistor of resistance $4R$ in series, such that $X_L = 3R$. Now, a capacitor is added in series such that $X_C = R$. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5} : x$ The value of x is _____.

(1) 1 (2) 2
(3) 3 (4) 4

16. An electron moving along the x-axis with an initial energy of 2.867 KeV enters a region of magnetic field $\vec{B} = 3.01\text{mT}\,\hat{k}$ at S (See figure). The field extends between $x = 0$ and $x = 3$ cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is (Electron's charge = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg) $\left(\sqrt{834.87} \cong 28.896\right)$



(1) 3.69 cm (2) 1.22 cm
(3) 11.65 cm (4) 2.25 cm

17. The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 500 nm is 0.62 V. When the incident wavelength is changed to a new value, the stopping potential is 1.86 V. The new wavelength is (approx.):

(1) 333 nm (2) 309 nm
(3) 429 nm (4) 400 nm

18. Hydrogen atom from n^{th} energy state comes to the ground state by emitting a photon of wavelength λ. The value of 'n' will be: (R: Rydberg constant)

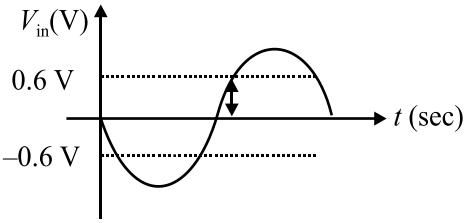
(1) $\sqrt{\frac{\lambda R}{\lambda R - 1}} + 1$ (2) $\sqrt{\frac{\lambda R}{\lambda R - 1}} - 1$

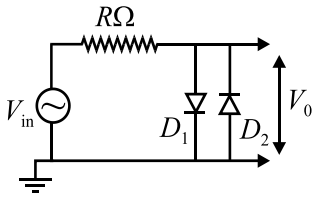
(3) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$ (4) $\sqrt{\frac{(\lambda R)^2}{\lambda R - 1}}$

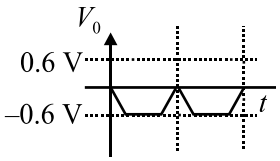
19. The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15})A^{1/3}$ m. The mass density of the nucleus is
($M_{\text{proton}} = M_{\text{neutron}} \simeq 1.67 \times 10^{-27}$ kg)

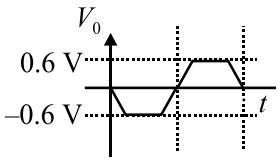
(1) 4.81×10^{17} kg m^{-3}
(2) 1.81×10^{17} kg m^{-3}
(3) 9.1×10^{17} kg m^{-3}
(4) 5.1×10^{18} kg m^{-3}

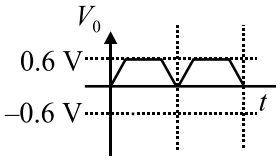
20. In the given circuit the input voltage V_{in} is shown in figure. The cut in voltage of *p*-n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diodes is correct?

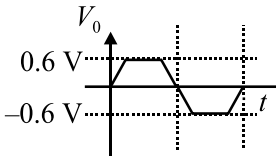




(1) 

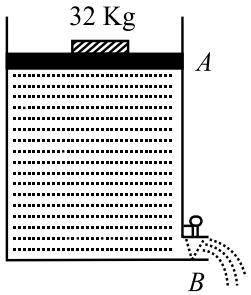
(2) 

(3) 

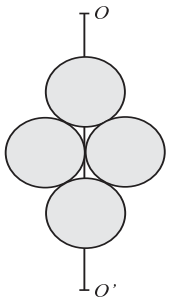
(4) 

Integer Type Questions

21. Consider a water tank as shown in the figure. Its cross-sectional area is 0.04 m^2. The tank has an opening *B* near the bottom whose cross-sectional area is 1 cm^2. A load of 32 kg is applied on the water at the top when the height of the water level is 45 cm above the bottom. The velocity of water coming out the opening *B* is_____ms^{-1}.
(Take value of *g* to be 10 ms^{-2}, density of water 1000 kg/m^3)

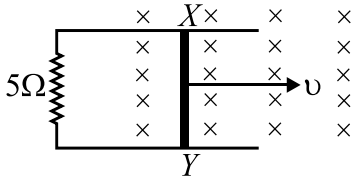


22. Four identical discs each of mass '*M*' and diameter '*a*' are arranged in a small plane as shown in figure. If the radius of gyration of the system about axis *OO*', in same plane as shown, is $\sqrt{x}\left(\frac{a}{4}\right)$. Then, the value of *x* will be

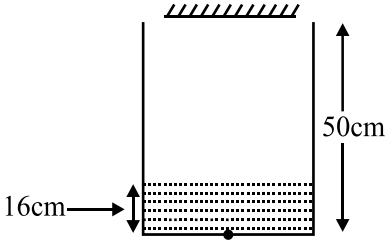


23. A parallel plate capacitor has plate area 7 m^2 and plate separation of 8 mm. The space between the plates is filled up to a thickness 5 mm with material of dielectric constant of 10. The resultant capacitance of the system is '*x*' nF. The value of '*x*' to the nearest integer is_____.
[ε_0= 8.85 × 10^{-12} Fm^{-1}]

24. A 0.5 m long metal rod *XY* completes the circuit as shown in figure. The plane of the circuit is perpendicular to the magnetic field of 0.25 T. If the resistance of the circuit is 5 Ω, the force needed to move the rod in direction, as indicated, with a constant speed of 8 m/s will be ____×10^{-3} N



25. As shown in the figure, a plane mirror is fixed at a height of 50 cm from the bottom of tank containing water $\left(\mu=\frac{4}{3}\right)$. The height of water in the tank is 16 cm. A small bulb is placed at the bottom of the water tank. The distance (in cm) of image of the bulb formed by mirror from the bottom of the tank is__.



SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. The mole fraction of KI in a 30% (w/w) aqueous solution is: [At. Mass: K = 39 u, I = 127 u]

(1) 0.125 (2) 0.081

(3) 0.044 (4) 0.451

27. The radius of the 5th Bohr orbit of Be^{3+}, in terms of the Bohr radius, a_0 is

(1) $4a_0$ (2) $\frac{25}{4}a_0$

(3) $\frac{4}{25}a_0$ (4) $\frac{4}{9}a_0$

28. The electronic configuration of two elements I & II are given as

$I \rightarrow 1s^2\ 2s^2\ 2p^3$

$II \rightarrow 1s^2\ 2s^2\ 2p^4$

The correct relation for their first ionisation potential (IP_1) is:

(1) IP_I of (I) = IP_I of (II)

(2) IP_I of (I) > IP_I of (II)

(3) IP_I of (I) < IP_I of (II)

(4) IP_I of (I) $\geq$ 2 times of IP_I of (II)

29. Molecular geometry and total number of lone-pairs of electrons of ClF_3 molecule are respectively.

(1) tetrahedral, 9

(2) octahedral, 10

(3) trigonal bipyramidal, 11

(4) T-shaped, 2

30. At 27°C, the value of K_p/K_c for the following reactions are respectively:

(At 27°C; RT = 24.62 L – atm mol^{-1})

(i) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(ii) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

(iii) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(1) 1; 0.0406 L^{-1} atm^{-1} mol; 24.62 L. atm. mol^{-1}

(2) 0.0406 L^{-1} atm^{-1} mol; 1; 24.62 L atm. mol^{-1}

(3) 1; 24.62 L.atm. mol^{-1}; 0.0406 L atm^{-1} mol

(4) 24.62 L. atm. mol^{-1}; 1; 0.0406 L^{-1} atm^{-1}mol

31. What is the oxidation state of Cr in CrO_5 and $[Cr(CO)_6]$ respectively.

(1) +2, +3

(2) +3, +3

(3) +6, 0

(4) +10, +6

32. If an element has the electron configuration $2s^2 2p^1$ in its valence shell, the electronic configuration of its diagonal element is:

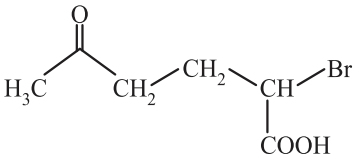
(1) [Ne] $3s^2\ 3p^2$

(2) [Ar] $4s^2\ 4p^2$

(3) [Ne] $4s^2\ 4p^2$

(4) [Ne] $2s^2\ 2p^2$

33. Which of the following is the correct IUPAC name of given compound?



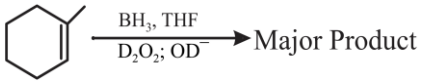
(1) 5–Formyl–2–bromohexanoic acid

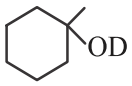
(2) 2–Bromo–5–oxohexanoic acid

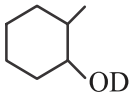
(3) 2–Formyl–5–bromohexan–6–oic acid

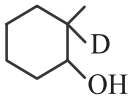
(4) 5–Bromo–2–oxohexan–6–oic acid

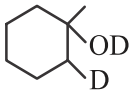
34. The major product obtained in the following reaction is/are:



(1) 

(2) 

(3) 

(4) 

35. In the process of separating a mixture of ethanol (I), ethoxyethane (II), and n-butanol (III) with a silica gel-packed chromatography column, a 75:25 hexane to ethyl acetate solvent mixture was used. The compounds were eluted in the following order:

(1) (I), (III) and (II)

(2) (I), (II) and (III)

(3) (II), (III) and (I)

(4) (II), (I) and (III)

36. For various half-cells, the $\frac{\Delta S}{nF}$ values are provided as follows:

I	II	III	IV
4×10^{-2}	3×10^{-2}	0.5×10^{-2}	0.3×10^{-2}

From the half-cells mentioned, which one is best suited to serve as a reference electrode?

(1) I (2) II

(3) III (4) IV

37. For the reaction X + Y → Z, if the reaction exhibits first-order kinetics with respect to each reactant X and Y, then what are the values of a and b respectively?

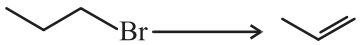
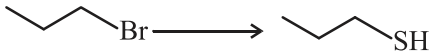
Rate $mol\,L^{-1}s^{-1}$	[X] $mol\,L^{-1}$	[Y] $mol\,L^{-1}$
0.20	10	0.25
0.80	a	0.25
1.60	5	b

(1) 40 and 4 (2) 8 and 40
(3) 20 and 4 (4) 2 and 5

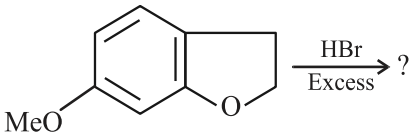
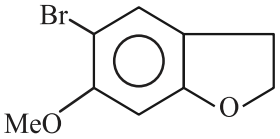
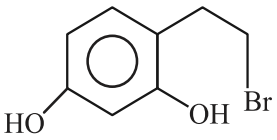
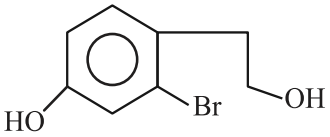
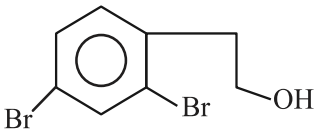
38. On heating, $Pb(NO_3)_2$ gives a brown gas "A". Which upon dimerization becomes colourless. What is the magnitude of change in the oxidation state of "N" in $Pb(NO_3)_2$ to Brown gas "A".
(1) 1 (2) 2
(3) 3 (4) 5

39. For a square planar complex $[Rh(py)(H_2O)(F)(NO_3)]$, the total number of isomers possible is;
(1) 2 (2) 6
(3) 8 (4) 3

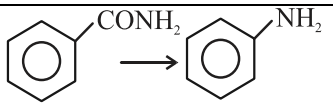
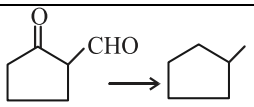
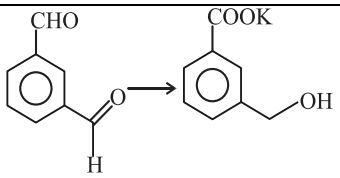
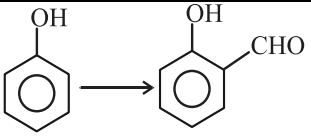
40. "A" is the neutral oxide of nitrogen but it is paramagnetic. When it reacts with O_2 it forms brown gas B. Gas "A" is absorbed by freshly prepared $FeSO_4$ solution, then what is "A"
(1) N_2O (2) NO
(3) NO_2 (4) N_2O_4

41. Which of the following reaction will take place at the fastest rate in presence of NaSH, ethanol at 25°C?
(1) 
(2) 
(3) 
(4) 

42. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion: 2, 4-dinitrophenol produces effervescence with $NaHCO_3$ while phenol doesn't.
Reason: Phenol is a very weak acid and 2, 4-nitrophenol is comparatively stronger acid.
In the light of the above statements, choose the correct answer from the options given below:
(1) A is false but R is true.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) Both A and R are true and R is the correct explanation of A.
(4) A is true but R is false.

43. Which of the following is the correct major product of the given reaction?

(1) 
(2) 
(3) 
(4) 

44. Match Column-I with Column-II

Column-I (Reaction)		Column-II (Reagents)	
A		P	Conc. KOH, Δ
B		Q	$CHCl_3$, NaOH/ H_3O^+
C		R	Br_2, NaOH
D		S	Zn–Hg/HCl

Choose the correct answer from the options given below:

	A	B	C	D
(1)	Q	S	P	R
(2)	R	S	Q	P
(3)	Q	P	R	S
(4)	R	S	P	Q

45. Given below are two statements, statement-I and statement-II.

Statement-I: When chlorobenzene reacts with sodium methoxide in presence of methanol, methyl phenyl ether formation takes place.

Statement-II: Nucleophilic aromatic substitution takes place in Williamson synthesis reaction of ethers.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is true but statement II is false
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is false but Statement II is true

Integer Type Questions

46. In a single-step adiabatic expansion of 2 moles of an ideal gas initially at 27°C, the work done by the gas is 2 kJ. The final temperature of the gas is_____ °C. [Nearest integer]
[Given: $C_v = 50\ Jmol^{-1}\ K^{-1}$]

47. A 100 L solution containing 20 g of NaOH was labelled as Solution X, and another 100 L solution containing 49 g of H_2SO_4 was labelled as solution Y. The pH of the resulting solution after mixing 80 L of Solution X with 20 L of solution Y is_____. [Nearest integer] (log 2 = 0.3)

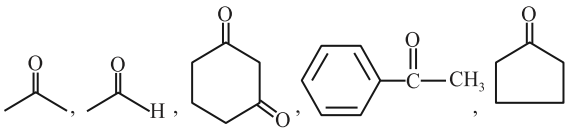
48. If the electrode potential for a half-cell reaction at pH = 9, standard atmospheric pressure of oxygen 1 bar and 298K is given as $x \times 10^{-1}$ volts. Then the value of x is _____. [Nearest integer]

$$2H^+(aq) + \frac{1}{2}O_2(g) + 2e^- \rightarrow H_2O(\ell); E^\circ = 1.23V$$

($R = 8.314\ J\ mol^{-1}\ K^{-1}$)

49. A compound 'A' gives violet color with $FeCl_3$ solution. Upon nitration with dilute HNO_3, it generates two isomers, (X) and (Y), which are separable by steam distillation. Reaction of compound 'A' with bromine water gives a white ppt of compound 'Z'. The combined total number of carbon, oxygen and bromine atoms present in compound 'Z' is_____.

50. How many of the following will show positive iodoform test.



SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5b^3c^2d$ is 6250 β, then the value of β is

(1) 54 (2) 90
(3) 27 (4) 108

52. The sum of squares of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is:

(1) $44+10\sqrt{3}$
(2) $45+18\sqrt{3}$
(3) $44-10\sqrt{3}$
(4) $45-18\sqrt{3}$

53. The equation $x^2 - 3x + [x] + 2 = x\,[x]$, where $[x]$ denotes the greatest integer function, has:

(1) Exactly two solutions in $(-\infty, \infty)$
(2) No solution
(3) A unique solution in $(-\infty, 1)$
(4) A unique solution in $(-\infty, \infty)$

54. The equation of line passing through mid-point of (3, 3, 4) and (–3, –7, 6) parallel to the line $\vec{r} = (\lambda - 1)\hat{i} + 2\hat{j} + 3\lambda\hat{k}$ is

(1) $\frac{x}{1} = \frac{y+2}{2} = \frac{z-5}{3}$
(2) $\frac{x}{1} = \frac{y-1}{2} = \frac{z}{3}$
(3) $\frac{x}{1} = \frac{y+2}{0} = \frac{z-5}{3}$
(4) $\frac{x}{1} = \frac{y-2}{0} = \frac{z}{3}$

55. A circle passes through the origin and has its centre on $y = x$. If it cuts $x^2 + y^2 - 4x - 6y + 10 = 0$ orthogonally, then the equation of the circle is

(1) $x^2 + y^2 - x - y = 0$
(2) $x^2 + y^2 - 6x - 4y = 0$
(3) $x^2 + y^2 - 2x - 2y = 0$
(4) $x^2 + y^2 + 2x + 2y = 0$

56. If for $z = \alpha + i\,\beta$, $|z + 2| = z + 5(1 + i)$, then $3\alpha + \beta$ and $3\alpha\beta$ are the roots of the equation

(1) $x^2 + 13x - 20 = 0$

(2) $x^2 + 13x + 30 = 0$

(3) $x^2 + x - 30 = 0$

(4) $x^2 + 11x + 30 = 0$

57. If the number of words, with or without meaning, which can be made using all the letters of the word VANCOUVER in which all the vowels do not come together, is $(6!)k$, then k is equal to

(1) 480 (2) 120

(3) 60 (4) 240

58. Let $f(x)$ = maximum $\{|x^2 - 2|x||, |x|\}$ and $g(x)$ = minimum $\{|x^2 - 2|x||, |x|\}$ then

(1) Both $f(x)$ and $g(x)$ are non differentiable at 5 points.

(2) $f(x)$ is not differentiable at 5 points and $g(x)$ is non differentiable at 7 points

(3) Number of points of non differentiability for $f(x)$ and $g(x)$ are 7 and 5 respectively

(4) Both $f(x)$ and $g(x)$ are non differentiable at 3 and 5 points respectively

59. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Then the relation $R = \{(x, y) \in A \times A : x + y = 7\}$ is

(1) Transitive but neither symmetric nor reflexive

(2) Reflexive but neither symmetric nor transitive

(3) An equivalence relation

(4) Symmetric but neither reflexive nor transitive

60. Let M be the maximum value of the product of two positive integers when their sum is 66. Let the sample space $S = \{x \in Z : x(75 - x) \geq \frac{6}{11}M\}$ and the event $A = \{x \in S : x$ is a multiple of $3\}$. Then probability $P(A)$ is equal to

(1) $\frac{9}{26}$ (2) $\frac{10}{29}$

(3) $\frac{13}{27}$ (4) $\frac{8}{27}$

61. If the numbers appeared on the two throws of fair six faced die are α and β, then the probability that $x^2 + 2\alpha x + \beta > 0$, for all $x \in R$, is:

(1) $\frac{7}{36}$ (2) $\frac{4}{9}$

(3) $\frac{1}{2}$ (4) $\frac{17}{36}$

62. Let $y = y(x)$ be a solution curve of the differential equation, $(1 - x^2y^2)dx = ydx + xdy$. If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 4$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α can be

(1) $\frac{5e^2}{2\left(5e^2 - 3\right)}$ (2) $\frac{5e^2 + 3}{2\left(5e^2 - 3\right)}$

(3) $\frac{-5e^2}{2\left(5e^2 + 3\right)}$ (4) $\frac{1 + 3e^2}{2\left(3e^2 - 1\right)}$

63. Let $A = \begin{bmatrix} 2 & 3 \\ \alpha & 0 \end{bmatrix}, \alpha \in R$ be written as $P + Q$ where P is a symmetric matrix and Q is skew symmetric matrix. If $\det(Q) = 36$, then the modulus of the sum of all possible values of determinant of P is equal to:

(1) 9 (2) 81

(3) 72 (4) 90

64. If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = 3/2$, then $\sin x + \cos y - \tan y$ is equal to

(1) $\frac{1 + \sqrt{3}}{2}$ (2) $\frac{1 - \sqrt{3}}{2}$

(3) $\frac{\sqrt{3}}{2}$ (4) $\sqrt{3}$

65. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Then the number of functions $f: A \to B$ satisfying $f(1) + f(2) = f(4) + 3$ is equal to

(1) 1080

(2) 1044

(3) 972

(4) 1008

66. $\lim\limits_{x \to 0}\left(\tan\left(\frac{\pi}{4} + x\right)\right)^{2/x}$ is equal to

(1) e^2 (2) e^{-2}

(3) e (4) e^4

67. Let f be a differentiable function in $\left(0, \frac{\pi}{2}\right)$. If $\int\limits_{\cos x}^{1} t^2 f(t)\,dt = \sin^3 x + \cos x - 1$, then $\left|f'\left(\frac{1}{\sqrt{5}}\right)\right|$ is equal to

(1) $\frac{5}{2}$ (2) $\frac{5\sqrt{5}}{2}$

(3) $\frac{2}{5\sqrt{5}}$ (4) $\frac{2}{\sqrt{5}}$

68. Let $f(x)=\int \frac{2x}{\left(x^2+1\right)\left(x^2+5\right)}dx$. If $f(2)=\frac{1}{4}\left(\log_e 5-\log_e 9\right)$, then $f(5)$ is equal to

(1) $\frac{1}{4}\left(\log_e 13-\log_e 15\right)$

(2) $\log_e 13-\log_e 15$

(3) $\frac{1}{4}\left(\log_e 15-\log_e 13\right)$

(4) $\log_e 15-\log_e 13$

69. The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{7x+\frac{\pi}{2}}{2-\cos 2x}dx$ is:

(1) $\frac{\pi^2}{6}$ (2) $\frac{\pi^2}{4\sqrt{3}}$

(3) $\frac{\pi^2}{6\sqrt{3}}$ (4) $\frac{\pi^2}{3\sqrt{3}}$

70. If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 4

S-I:- Length of focal chord is 31

S-II: Slope of focal chord is $-\frac{2}{3}$

(1) S-I is true, S-II is false

(2) S-I is false, S-II is true

(3) Both S-I and S-II are true

(4) Both S-I and S-II are false

Integer Type Questions

71. If the area of the region, given by the set $\left\{(x,y)\in R\times R,\ x\geq 0,\ x^2\leq y\leq 2-x\right\}$ is $\frac{p}{q}$ (where p, q are co-primes), then the value of $p + q$ is

72. A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x-axis, y-axis respectively. If the perpendicular from Origin O to line makes an angle of $\frac{\pi}{3}$ with positive direction of y-axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then $\frac{b^2}{a^2}$ is equal to

73. Let $x = \sin(2\tan^{-1}\alpha)$ and $y=\sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \{\alpha \in R : y^2 = 1 - x\}$, then $\sum_{\alpha\in S} 8\alpha^3$ is equal to

74. Let $\vec{a}=6\hat{i}+9\hat{j}+12\hat{k}$, $\vec{b}=\alpha\hat{i}+11\hat{j}-2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a}\times\vec{c}=\vec{a}\times\vec{b}$. If $\vec{a}\cdot\vec{c}=-12$, $\vec{c}\cdot\left(\hat{i}-2\hat{j}+\hat{k}\right)=5$, then $\vec{c}\cdot\left(2\hat{i}+\hat{j}+2\hat{k}\right)$ is equal to

75. The mean and variance of 5 observations are 5 and 8 respectively. If 3 observations are 1, 3, 5, then the absolute value of difference of cubes of the remaining two observations is

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